

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

TCP/IP layer	Example Packet	What to look for in Wireshark	Example Header Fields
Application	HTTP2 or TLS	Protocol column shows HTTP2, TLSv1.2 and QUIC	Host, URI, Content-Type and Server Name
Transport	TCP	Protocol column shows TCP	Source Port, Destination Port(443), Sequence Number and Acknowledgment Number
Internet	IPv4	Protocol column shows IPv4	Source IP, Destination IP, TTL, Protocol=TCP
Network Access	Ethernet II	Protocol column shows Ethernet II	Source MAC Address, Destination MAC Address, EtherType = IPv4

- **Application Layer Packet**
- Protocol : TLS 1.2
- It establishes a secure encrypted connection with Flipkart.
- Need to check: Server Name, TLS Version.
- **Transport Layer Packet**
- Protocol : TCP
- It provides reliable communication between my computer and Flipkart server.
- Need to check: Source Port, Destination Port (443) and Sequence Number.

- **Internet Layer Packet**
- Protocol : IPv4
- It routes packets across the Internet.
- Need to check: Source IP Address, Destination IP Address and TTL.
- **Network Access Layer Packet**
- Protocol : Ethernet II
- It delivers frames between my computer and the local router.
- Need to check: Source MAC address, Destination MAC address and EtherType.

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI Model Layer	TCP/IP Model Layer	Real-World Example
1. Application	Application	Sending a WhatsApp message.
2. Presentation	Application	WhatsApp encrypts my message and compress it before sending.
3. Session	Application	WhatsApp keeps chat session active during continue messaging.
4. Transport	Transport	TCP ensures WhatsApp message sends it correctly.
5. Network	Internet	IP finds the route from phone to WhatsApp server over the internet.
6. Data-link	Network Access	Phone sends Wi-Fi frames to the router using its MAC address.
7. Physical	Network Access	Data is transmitted as electrical signals using Ethernet and Wi-Fi.

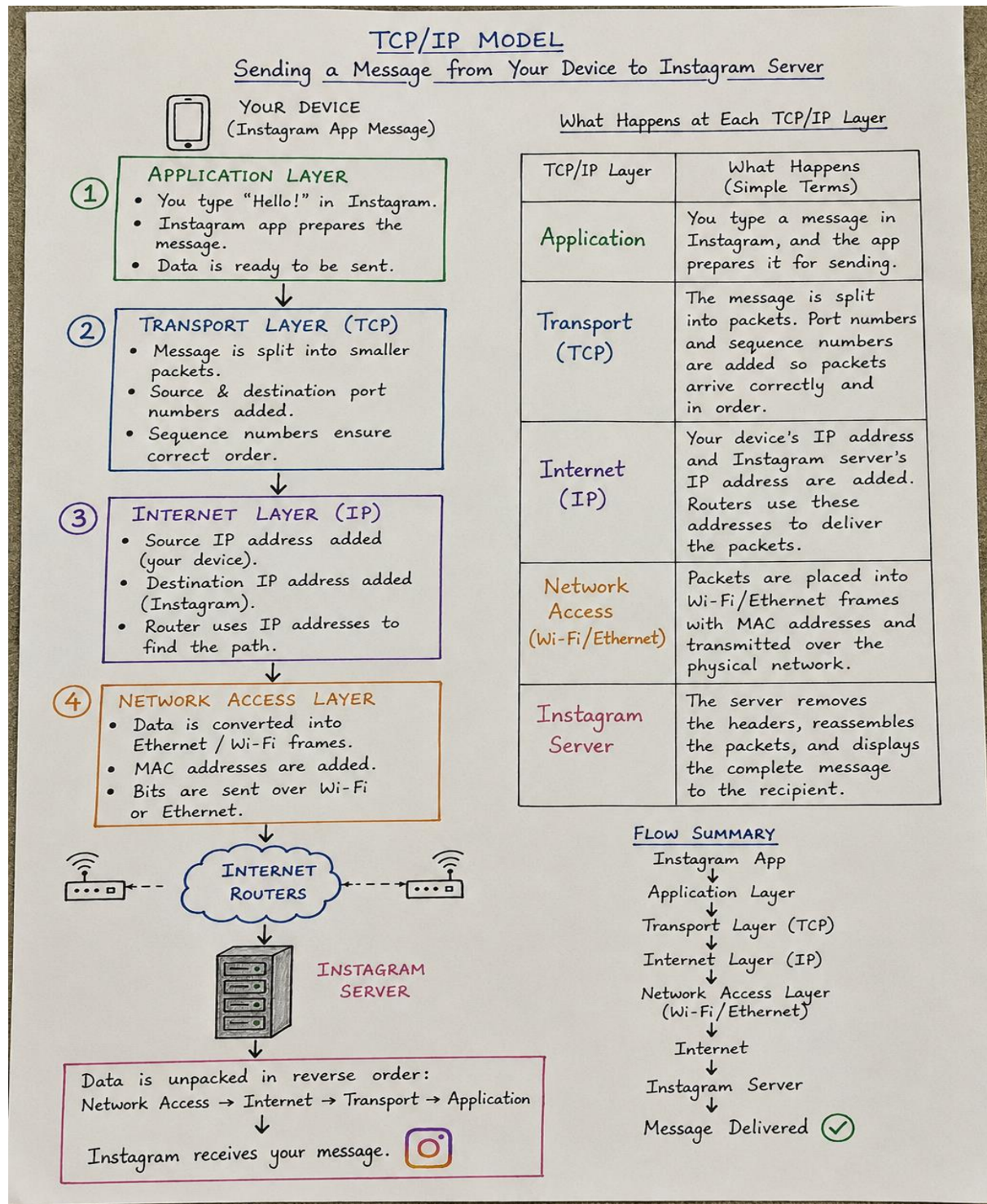
OSI Model	TCP/IP Model
It has 7 layers.	It has 4 layers.
It is theoretical model used for the learning and troubleshooting.	A Practical model used on the internet.
It has three separate layers which are Application, Presentation and Session.	It combines all three layers in the one layer which is Application layer.
It has two separate layers which are Data-Link and Physical.	It combines both layers in the one which is Network Access Layer.

3. Given a screenshot of a TCP segment header, label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

- These are the Transport layer fields need to check in TCP segment header.

TCP Header	Purpose
Source Port	Identifies the application or service that sent the data (ex. Web browser using a temporary port).
Destination Port	Identifies the application or service that should receive the data.
Sequence Port	Specifies the number of the first byte of data in this segment. It helps keep data in the correct order.
Acknowledgement Port	Indicates the next byte the receiver expects to receive, confirming that previous data arrived successfully.
Header Length	Shows the size of the TCP header so the receiver knows where the actual data begins.
Reserved	Reserved for future use and normally set to 0.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

- Here, I have selected TCP Packet.
- I selected a TCP packet from my Wireshark capture. It belongs to the Transport Layer of the TCP/IP model. I identified it because the Protocol column displayed TCP, and the packet contained TCP header fields such as Source Port, Destination Port, Sequence Number and Acknowledgement Number. In a real-world application, this packet represents the reliable transfer of data, such as when a WhatsApp message is sent from one user to another, ensuring the message reaches the destination correctly and in order.